KUNDAN KUMAR

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EDUCATION

Wayne State University, Detroit, MI Dec 2024
Master of Science (M.S.) - Data Science and Business Analytics

Relevant Coursework: Data Science & Analytics, Computing Platform for Data Science, Machine Learning, Deep Learning, Statistical Learning for Data Science, Decision Analysis & Simulation, Data Mining, Operations Research

WORK EXPERIENCE

Henry Ford Health, Detroit, MI, USA

Analyst – Experience Analytics Feb 2025 – Current

- Deployed advanced **SQL** techniques to retrieve and shape data solutions from **Data Warehousing (Azure),** ensuring precise analytics; accelerated data extraction by 35% and improved dataset relevancy by 25%.
- Analyzed healthcare satisfaction data from **Press Ganey**, delivering insights that helped clinicians and physicians implement changes resulting in a 22% improvement in patient feedback scores.
- Created and launched interactive healthcare dashboards in **Power BI**, revealing trends and key performance metrics that boosted decision-making efficiency by 40% and reduced report generation time by 60%.
- Collaborated with leadership to develop and implement the CMS-mandated **Psychiatric Inpatient Experience** (PIX) Survey in alignment with IPFQR program requirements.

Intern/Co-Op Student, CX Insights & Analytics May 2024 – Feb 2025

- Conducted performance analysis of healthcare providers using Press Ganey data and analyzed patient data on **EMR (Electronic Medical Record)** reports in **Epic Systems.**
- Ensured accuracy, thoroughness, and **attention to detail** in all data analysis tasks.
- Demonstrated active listening and provided constructive feedback to team members.

IndustryX, Detroit, MI, USA May 2024 – Jul 2024

Data Science in Robotics (Academic Practicum Internship)

Project: Automated Disassembly Process Planning with Vision and AI

- **Fastener Identification using Machine Learning:**
  - Developed **Machine Learning** solutions to detect fasteners in visual data with 94% accuracy, improving inventory management and lowering identification errors by 25%.
  - Fine-tuned pre-trained Yolov8 models using **Python**, achieving 95% accuracy in fastener detection and reducing manual inspection time by 60%, boosting overall operational efficiency.
  - Conducted comprehensive model testing in Nvidia's ISAAC SIM, identifying and rectifying 15 critical performance issues, improving simulation fidelity by 35%.
- **3D Geometry Capture via Camera/LIDAR:**
  - Implemented robot-mounted camera and LIDAR-based scanning systems to capture detailed 3D part geometry, leveraging ISAAC SIM for realistic simulation scenarios.
  - Conducted development and testing in a virtual environment, leveraging ISAAC SIM for realistic simulation scenarios.

Eve Healthcare Centers Private Limited, U.P, India Jun 2022 - Dec 2022

Machine Learning Intern

- Collaborated with ~4 national cross-functional teams to develop innovative product solutions for Eve Healthcare Centers, utilizing **Machine Learning** methodologies to deliver data-driven insights for +50k customers.
- Scaled a Channel Preference ML Model building to analyze customer reaction towards offers from different channels. Optimized this model from 25 hrs to 5 hrs (80% reduction) using **Azure** and **Kubernetes** friendly practices.
- Showcased attention to detail and timeliness in all project deliveries.

5ONE Analytics, **Haryana, India** Jan 2021 - May 2022

Data Scientist

- Validated data, conducted **data mapping**, and modeled diverse features for AI/ML applications; extracted historical data from the **Enterprise Data Warehouse** (EDW) using **AWS Glue** and created hierarchical agent and branch structures with **PySpark,** meeting latency requirements of 1 millisecond.
- Excelled in presenting information, conflict resolution, and communication within teams.

PROJECTS

Market Due Diligence and Asset Identification for Pharmaceutical: (GitHub)

Indegene – Analytics Case Study using Python

- Analyzed datasets after data gathering provided by the client and performed additional research using public data sources.
- Identified attractive therapeutic areas (ATC 3) for investment and recommended high-potential and top-priority molecules (ATC 4) for product acquisition.
- Projected the company's revenue growth from 267 million USD in 2018 to 2023, considering a historical CAGR of 5%.

Predictive Analysis and Dynamic Visualization of Chronic Disease Trends in the USA: A Flask-based Application on AWS: (GitHub)

- Led a process flow and thorough **Exploratory Data Analysis (EDA)** on a multifaceted chronic disease dataset, adeptly managing data cleaning processes and extracting insights for trend identification.
- Engineered and deployed a dynamic Flask-based application on AWS, featuring predictive models (Linear Regression, XGBOOST, Random Forest) to forecast alcohol use trends among youth, exemplifying proficiency in analytics and deployment with a measurable impact on public health insights.

TECHNICAL SKILLS

- **Data Analysis Tools and Languages:** Python, SQL, R
- **Business Intelligence Tools & Databases:** Statistics, Tableau, PowerBI, Jupyter Notebook, Microsoft Excel, PowerPivot, flask, CI/CD Pipeline, Relational Database (MS SQL Server),
- **Machine Learning Techniques:** Classification, Regression, Time Series, Anomaly Detection, Decision Trees, Probability networks, Natural Language Processing, Association Rules, Clustering, Neural Networks, Bayesian models, LLMS, Generative AI
- **Cloud Competencies:** AWS, GCP, Microsoft Azure

CERTIFICATIONS & ACCOMPLISHMENTS

- **Python Masterclass by Tim Buchalka by** *Udemy 2020*
- **Data Visualization with Tableau Specialization by** *University of California Davis, Coursera 2023*
- **The complete SQL masterclass, by** *Udemy*
- **The complete Microsoft PowerBI Bootcamp, By** *Udemy*